

Sets

1) Unordered & Unindexed collection of items.

2) Set elements are unique. Duplicate elements are not allowed.

3) Set elements are immutable (cannot be changed).

4) Set itself is mutable. We can add or remove items from it.

```
In [4]: myset = {1,2,3,4,5} # Set of numbers  
myset
```

```
Out[4]: {1, 2, 3, 4, 5}
```

```
In [5]: len(myset)
```

```
Out[5]: 5
```

```
In [6]: my_set = {1,1,2,2,3,4,5,5}  
my_set
```

```
Out[6]: {1, 2, 3, 4, 5}
```

```
In [7]: myset1 = {1.79,2.08,3.99,4.56,5.45} # Set of float numbers  
myset1
```

```
Out[7]: {1.79, 2.08, 3.99, 4.56, 5.45}
```

```
In [8]: myset2 = {'Asif' , 'John' , 'Tyrion'} # Set of Strings  
myset2
```

```
Out[8]: {'Asif', 'John', 'Tyrion'}
```

```
In [9]: myset3 = {10,20, "Hola", (11, 22, 32)} # Mixed datatypes  
myset3
```

```
Out[9]: {(11, 22, 32), 10, 20, 'Hola'}
```

```
In [10]: myset3 = {10,20, "Hola", [11, 22, 32]} # set doesn't allow mutable items like li  
myset3
```

```
-----  
TypeError                                Traceback (most recent call last)  
Cell In[10], line 1  
----> 1 myset3 = {10,20, "Hola", [11, 22, 32]} # set doesn't allow mutable items  
like li  
      2 myset3  
  
TypeError: unhashable type: 'list'
```

```
In [11]: myset4 = set() # Create an empty set
print(type(myset4))
```

```
<class 'set'>
```

```
In [13]: my_set1 = set(('one' , 'two' , 'three' , 'four'))
my_set1
```

```
Out[13]: {'four', 'one', 'three', 'two'}
```

Add & Remove Items

```
In [14]: myset
```

```
Out[14]: {1, 2, 3, 4, 5}
```

```
In [15]: myset.add('NINE') # Add item to a set using add() method
myset
```

```
Out[15]: {1, 2, 3, 4, 5, 'NINE'}
```

```
In [16]: myset = {'one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight'}
for i in myset:
    print(i)
```

```
one
seven
five
four
three
eight
two
six
```

```
In [17]: myset.add('NINE') # Add item to a set using add() method
myset
```

```
Out[17]: {'NINE', 'eight', 'five', 'four', 'one', 'seven', 'six', 'three', 'two'}
```

```
In [18]: myset.update(['ten', 'eleven', 'twelve'])
myset
```

```
Out[18]: {'NINE',
'eight',
'eleven',
'five',
'four',
'one',
'seven',
'six',
'ten',
'three',
'twelve',
'two'}
```

```
In [19]: myset.remove('NINE')
myset
```

```
Out[19]: {'eight',
          'eleven',
          'five',
          'four',
          'one',
          'seven',
          'six',
          'ten',
          'three',
          'twelve',
          'two'}
```

```
In [20]: myset.discard('ten')
myset
```

```
Out[20]: {'eight',
          'eleven',
          'five',
          'four',
          'one',
          'seven',
          'six',
          'three',
          'twelve',
          'two'}
```

```
In [21]: myset.clear()
```

```
In [22]: del myset
```

```
In [23]: myset = {'one', 'two', 'three', 'four', 'five', 'six', 'seven', 'eight'}
myset
```

```
Out[23]: {'eight', 'five', 'four', 'one', 'seven', 'six', 'three', 'two'}
```

```
In [24]: myset1 = myset # Create a new reference "myset1"
myset1
```

```
Out[24]: {'eight', 'five', 'four', 'one', 'seven', 'six', 'three', 'two'}
```

```
In [25]: my_set = myset.copy() # Create a copy of the List
my_set
```

```
Out[25]: {'eight', 'five', 'four', 'one', 'seven', 'six', 'three', 'two'}
```

```
In [27]: myset.add('nine')
myset
```

```
Out[27]: {'eight', 'five', 'four', 'nine', 'one', 'seven', 'six', 'three', 'two'}
```

```
In [28]: myset1
```

```
Out[28]: {'eight', 'five', 'four', 'nine', 'one', 'seven', 'six', 'three', 'two'}
```

```
In [29]: my_set
```

```
Out[29]: {'eight', 'five', 'four', 'one', 'seven', 'six', 'three', 'two'}
```

Set operation

```
In [30]: A = {1,2,3,4,5}
        B = {4,5,6,7,8}
        C = {8,9,10}
```

```
In [31]: A | B
```

```
Out[31]: {1, 2, 3, 4, 5, 6, 7, 8}
```

```
In [33]: A|(B|C)
```

```
Out[33]: {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}
```

```
In [34]: A.update(B,C)
        A
```

```
Out[34]: {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}
```

```
In [35]: # Intersection
```

```
In [37]: A = {1,2,3,4,5}
        B = {4,5,6,7,8}
```

```
In [38]: A & B
```

```
Out[38]: {4, 5}
```

```
In [39]: A.intersection(B)
```

```
Out[39]: {4, 5}
```

```
In [40]: A.intersection_update(B)
```

```
In [41]: A
```

```
Out[41]: {4, 5}
```

```
In [42]: A = {1,2,3,4,5}
        B = {4,5,6,7,8}
```

```
In [43]: A - B
```

```
Out[43]: {1, 2, 3}
```

```
In [44]: A.difference(B)
```

```
Out[44]: {1, 2, 3}
```

```
In [45]: B - A
```

```
Out[45]: {6, 7, 8}
```

In [46]: `A ^ B`

Out[46]: `{1, 2, 3, 6, 7, 8}`

In [47]: `A = {1,2,3,4,5,6,7,8,9}`
`B = {3,4,5,6,7,8}`
`C = {10,20,30,40}`

In [48]: `B.issubset(A)`

Out[48]: `True`

In [49]: `C.issubset(A)`

Out[49]: `False`

In [50]: `A.issuperset(B)`

Out[50]: `True`

In [51]: `C.issuperset(B)`

Out[51]: `False`

In [52]: `C.isdisjoint(A)`

Out[52]: `True`

In [53]: `B.isdisjoint(A)`

Out[53]: `False`

In []: